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“

Branding is
what people
say about
you when
you are not
in the room.

”



-Jeff Bezos



```

lastUpdate = millis();

float co = sensorConnected ? sensor.measureCO() : 0;
float no2 = sensorConnected ? sensor.measureNO2() : 0;
float nh3 = sensorConnected ? sensor.measureNH3() : 0;
float c3h8 = sensorConnected ? sensor.measureC3H8() : 0;
float c4h10 = sensorConnected ? sensor.measureC4H10() : 0;
float ch4 = sensorConnected ? sensor.measureCH4() : 0;
float h2 = sensorConnected ? sensor.measureH2() : 0;
float c2h5oh = sensorConnected ? sensor.measureC2H5OH() : 0;

// Trigger BME measurement
bme69x_set_op_mode(BME69X_FORCED_MODE, &dev);
dev.delay_us(100000, dev.intf_ptr);

float temp = 0, press = 0, hum = 0, gas = 0;
uint8_t n_fields;
if (bme69x_get_data(BME69X_FORCED_MODE, &data, &n_fields, &dev) == BME69X_OK && n_fields > 0) {
    temp = data.temperature;
    press = data.pressure / 100.0;
    hum = data.humidity;
    gas = data.gas_resistance;
}

float esptemp = temperatureRead();
int clients = WiFi.softAPgetStationNum();
String timestamp = timeClient.getFormattedTime();

// Build JSON
String json = "{";
json += "\"timestamp\": \"" + timestamp + "\", ";
json += "\"co\": " + String(co, 2) + ", ";
json += "\"no2\": " + String(no2, 2) + ", ";
json += "\"nh3\": " + String(nh3, 2) + ", ";
json += "\"c3h8\": " + String(c3h8, 2) + ", ";
json += "\"c4h10\": " + String(c4h10, 2) + ", ";
json += "\"ch4\": " + String(ch4, 2) + ", ";
json += "\"h2\": " + String(h2, 2) + ", ";
}

```

Fig. 3: Code snippet function to update sensor data in real time

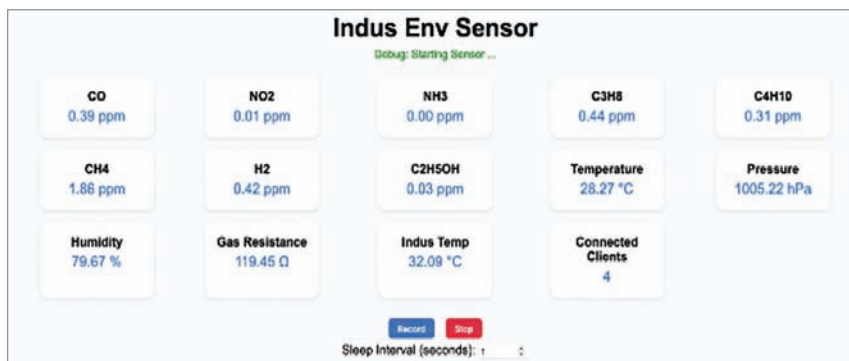


Fig. 4: Webpage UI of the monitor sensor

every second in real time.

A webpage UI dashboard displays real-time data and allows CSV recording and download. Fig. 2 shows the code snippet that configures the Wi-Fi SSID and password, while Fig. 3 shows the function for updating sensor data in real time. Fig. 4 shows the webpage UI of the monitor sensor.

Circuit diagram

Fig. 5 shows the circuit diagram of the IoT monitor sensor. It is built

around an IndusBoard Coin V2 (U1), BME690 (U2), MICS6814 (U3), GC9A01 LCD (LCD1), and a few other components. A 3.3V or 3.7V battery powers the device. A 24-pin, 0.5mm FPC ribbon cable interconnects the components.

This customised sensor module is designed for easy integration with the IndusBoard Coin but is also compatible with other ESP or similar development boards. The gas and environmental sensors are connected to the I²C pins (default